Problem 7 – ProofCouncil

Abstract

Let f be a quadratic rational map one of whose critical points lies in a cycle A of length four and whose other critical value v lies outside A, and let $B = A \cup \{v\}$. Epstein's deformation space $\mathcal{D}_f \subset \mathcal{T}(\mathbb{S}^2, B)$ is known to be disconnected (Hironaka–Koch). We write down an explicit simple closed curve γ on $(\mathbb{S}^2, B)$, for a normalized model map the image of a round circle under two Dehn twists about a second round circle, transported to a general f along a path in parameter space, and prove that the Dehn twist T_γ itself preserves $\mathcal{D}_f$ and that the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the base component D_0 are pairwise distinct components, using the covering-space framework of Hironaka–Koch and a winding-number obstruction.

1 Introduction

1.1 Deformation spaces

Throughout, $\mathbb{P}^1$ is the Riemann sphere, which we also regard as the oriented topological sphere $\mathbb{S}^2$. For a finite set $B \subset \mathbb{P}^1$ with $|B| \geq 3$, the *Teichmüller space* $\mathcal{T}(B) = \mathcal{T}(\mathbb{S}^2, B)$ is the set of orientation-preserving homeomorphisms $\phi\colon \mathbb{P}^1 \to \mathbb{P}^1$ (*markings*), where $\phi \sim \phi'$ when $\phi' = \mu \circ \phi \circ h$ for a Möbius transformation μ and a homeomorphism h isotopic to the identity relative to B. Pulling back the standard complex structure by ϕ identifies $\mathcal{T}(B)$ with the complex structures on $(\mathbb{S}^2, B)$ up to isotopy relative to B; the class $\tau_0 = [\mathrm{id}]$ is the *base point*.

Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be an orientation-preserving branched cover of degree $d \geq 2$ with set of critical values $V(f)$, and let $A, B \subset \mathbb{P}^1$ be finite sets with $|A|, |B| \geq 3$ and $f(A) \cup V(f) \subset B$. Pulling back the complex structure of a marking ϕ under f gives a complex structure on $(\mathbb{S}^2, A)$; concretely, there are a homeomorphism ψ, unique up to post-composition with a Möbius transformation, and a rational map F with

$$F \circ \psi = \phi \circ f, \tag{1.1}$$

and $f^*[\phi] := [\psi] \in \mathcal{T}(A)$. This is the Thurston pullback map $f^*\colon \mathcal{T}(B) \to \mathcal{T}(A)$; it is holomorphic [BCT, §1.3, Prop. 1.4]. We call ϕ the *target marking* and ψ the *source marking* of the diagram (1.1). If $A \subset B$, forgetting the points of $B \setminus A$ gives the holomorphic *forgetful*

map $i_*\colon \mathcal{T}(B) \to \mathcal{T}(A)$, $[\phi] \mapsto [\phi]$. When $f(A) \cup V(f) \cup A \subset B$, both maps are defined, and the *deformation space* of f is their equalizer

$$\mathcal{D}_f = \{\tau \in \mathcal{T}(B) : f^*(\tau) = i_*(\tau)\}.$$

This notion is due to Epstein [Ep]; see [BCT, §1.4] for an exposition. When f is rational, as in the problem considered here, we may take $\psi = \mathrm{id}$, $F = f$ in (1.1) for $\phi = \mathrm{id}$, so $\tau_0 \in \mathcal{D}_f$. (By Epstein's theorem [BCT, Thm. 1.5], under the standing hypotheses above, if f is not a topological $(2,2,2,2)$-map in the sense of [BCT, §1.4], then $\mathcal{D}_f$, whenever nonempty, is a smooth complex submanifold of $\mathcal{T}(B)$ of dimension $|B \setminus A|$. For the rational family considered here $\tau_0 \in \mathcal{D}_f$, and the periodic critical point excludes the exceptional type.)

The *pure mapping class group* $\mathrm{PMod}(B)$, the group of isotopy classes relative to B of orientation-preserving homeomorphisms of $\mathbb{P}^1$ fixing B pointwise, acts on $\mathcal{T}(B)$ on the right by

$$[\phi] \cdot h = [\phi \circ h]; \tag{1.2}$$

for a subset $D \subset \mathcal{T}(B)$ we write $h(D) = \{\tau \cdot h : \tau \in D\}$. A mapping class that preserves $\mathcal{D}_f$ permutes its connected components. For a simple closed curve γ in $\mathbb{P}^1 \setminus B$ we write $T_\gamma \in \mathrm{PMod}(B)$ for the Dehn twist about γ. The question addressed here asks for an explicit γ such that some power of T_γ preserves $\mathcal{D}_f$ and has an infinite orbit on the components.

1.2 The family and the result

Let f be a quadratic rational map with critical points c_1, c_2 such that c_1 lies in a cycle of length 4 and the critical value $v = f(c_2)$ is not in this cycle, and put

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \qquad B = A \cup \{v\}.$$

Then $f(A) = A$ and $V(f) = \{f(c_1), v\} \subset B$, so $\mathcal{D}_f \subset \mathcal{T}(B)$ is defined; $\mathcal{T}(B)$ has dimension 2. Conjugating f by the Möbius transformation that sends $c_1, f(c_1), f^{\circ 2}(c_1)$ to $0, \infty, 1$ normalizes the cycle to $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$, and Section 3 shows that this forces f to become

$$F_s(z) = 1 + \frac{-s^2 - s + 1}{(s-1)z} + \frac{s(2s-1)}{(s-1)z^2}, \qquad s \in U, \tag{1.3}$$

for an explicit Zariski-open set $U \subset \mathbb{C}$ given in (3.2). Our model is F_a with $a = 3/5$, whose marked set is $B_a = \{0, \infty, 1, a, v_a\}$, where $v_a = 121/120$ is its second critical value (computed in (3.1)).

The answer (Theorem 3.2) is as follows. Let C be the round circle $|z| = 241/240$, which separates $\{\infty, v_a\}$ from $\{0, 1, a\}$, and let η be the round circle $|z - 4/5| = 1/4$, which encloses exactly $a, 1, v_a$. Put $\gamma_a = T_\eta^2(C)$, the image of C under two Dehn twists about η (with the sign of Convention 2.1; either sign works, Remark 7.4). For a general f of the family, let γ be obtained from γ_a by following any path in U from a to the parameter of f and dragging the marked points along. Then T_γ preserves $\mathcal{D}_f$, and the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the component $D_0 \ni \tau_0$ are pairwise distinct components; so the power of the twist can be taken to be 1. (A path is needed: γ_a and C separate B_a in the same way, so separation data do not determine the isotopy class of γ.)

1.3 Relation to earlier work

Hironaka and Koch [HK] proved that for this family $\mathcal{D}_f$ has infinitely many connected components ([HK, Thm. 1.4]), introducing the tools we use: the surface of marked quadratic maps with its degree-four covering of moduli space and its diagonal ([HKa, §§4.1–4.2], [HK, §3]), the description of $\mathcal{D}_f$ through mapping classes that lift under f and through equalizers ([HK, §§2.8–2.11]), and a symmetric quotient of the parameter surface used as an obstruction ([HK, §2.12]); they also exhibit a mapping class preserving $\mathcal{D}_f$ with an infinite orbit on components ([HK, Prop. 3.2]). Here [HKa] is the first arXiv version of their paper and [HK] the revised author version, whose numbering we follow. Firsova, Kahn and Selinger [FKS] treat the same family ([FKS, §6]; numbering of arXiv v4), study twists along multicurves adapted to the pullback ("equalizing multicurves"), and prove that these deformation spaces are not contractible ([FKS, Lemma 8.4]). The present note adds the identification of a single, explicitly drawn Dehn twist with an infinite component orbit, which we have not found in [HK] or [FKS], with an elementary verification; our obstruction map Φ (Section 7) is a form of the symmetric quotient of [HK].

1.4 Strategy and outline

The proof has five steps. (1) Section 4 factors the universal cover $\mathcal{T}(B_a) \to M$ of the moduli space M of five marked points through a finite cover $\mathcal{W} \to M$, the surface of quadratic maps with the prescribed portrait of marked orbits; the pullback and forgetful maps become the coordinate projections of $\mathcal{W}$, and $\mathcal{D}_{F_a}$ lies over the diagonal $\mathcal{V} \subset \mathcal{W}$. (2) Section 5: a loop g in $\mathcal{W}$ acts on $\mathcal{T}(B_a)$ as a mapping class, which preserves $\mathcal{D}_{F_a}$ if the two coordinate loops of g are homotopic, and preserves the base component D_0 if and only if g is homotopic in $\mathcal{W}$ to a loop in $\mathcal{V}$. (3) Section 6 exhibits a loop $g = \ell^{-1} m \ell$ in $\mathcal{W}$ acting as $T_{\gamma_a}^{\pm 1}$: m is a meridian acting as the twist about C, and the point push along ℓ is identified exactly by a Joukowski substitution. (4) Section 7 shows that no power g^n, $n \neq 0$, is homotopic to a loop in $\mathcal{V}$, using a map $\Phi = (t, L) \colon \mathcal{W} \to (\mathbb{C} \setminus \{0, 1\}) \times \mathbb{C}^*$ whose first coordinate becomes, on an enlarged diagonal, the two-sheeted covering $w \mapsto w^2$, while its second coordinate imposes a linear relation between two winding numbers that the loops g^n violate. (5) Section 8 transports the conclusion to every map of the family by lifting an isotopy of marked points through the branched covers.

2 Conventions on loops, point pushing and twists

For B as above, the *moduli space* $\mathcal{M}(B)$ is the set of injective maps $B \to \mathbb{P}^1$ modulo post-composition with Möbius transformations. Choosing three points of B and normalizing their images to $0, \infty, 1$ identifies $\mathcal{M}(B)$ with an open subset of $\mathbb{C}^{|B|-3}$. The projection $\pi_B \colon \mathcal{T}(B) \to \mathcal{M}(B)$, $[\phi] \mapsto [\phi|_B]$, is a holomorphic universal covering whose deck group is $\mathrm{PMod}(B)$ acting by (1.2); see [KPS, §2.1].

Two moduli spaces occur. For four marked points labelled $0, \infty, 1, x$, and for five points

labelled $0, \infty, 1, y, z$, we use respectively

$$Y = \mathbb{C} \setminus \{0, 1\}, \qquad M = \{(y, z) \in Y^2 : y \neq z\}. \tag{2.1}$$

Thus for $A \supset \{0, \infty, 1\}$ with $|A| = 4$, $\pi_A \colon \mathcal{T}(A) \to Y$ records the fourth point under the marking normalized to fix $0, \infty, 1$, and similarly $\pi_B \colon \mathcal{T}(B) \to M$ when $|B| = 5$.

Products of based loops are read from right to left: $\alpha\beta$ traverses β first and then α, matching composition of homeomorphisms. For a closed loop ω in $\mathbb{C}\setminus\{c\}$ we write $\mathrm{wind}_c(\omega) \in \mathbb{Z}$ for its winding number about c; for a non-closed path, $\mathrm{wind}_c(\omega)$ denotes the total change of argument of $\omega - c$ divided by 2π, which is additive under concatenation and changes sign under reversal.

Let δ be a loop in $\mathcal{M}(B)$ based at $\pi_B(\tau_0)$, that is, a motion of the marked points. By isotopy extension there is an ambient isotopy h_t of $\mathbb{P}^1$ with $h_0 = \mathrm{id}$ such that $h_t|_B$ follows $\delta(t)$. The mapping class $P(\delta) = [h_1] \in \mathrm{PMod}(B)$ depends only on the homotopy class of δ, and $[h_t]$ is the lift of δ to $\mathcal{T}(B)$ starting at τ_0, with endpoint $\tau_0 \cdot P(\delta)$. Consequently the deck transformation associated with δ, the one sending τ_0 to the endpoint of this lift, is

$$[\phi] \longmapsto [\phi \circ P(\delta)]. \tag{2.2}$$

Point pushing respects products, $P(\alpha\beta) = P(\alpha)P(\beta)$, since the isotopy following $\alpha\beta$ may be taken to be h_t^{β} followed by $h_t^{\alpha} \circ h_1^{\beta}$. Because (2.2) is a right action, deck transformations of products compose in the opposite order; we only use deck transformations of powers of a single loop. We write $g(\tau)$ for the deck transformation (2.2) of a loop g applied to τ.

When only one marked point v moves, δ is a loop ζ in $\mathbb{P}^1 \setminus (B \setminus \{v\})$ based at v, and $P(\zeta)$ is the *point-pushing map* $\mathrm{Push}(\zeta)$ of the Birman exact sequence [FM, §4.2, Thm. 4.6]. We use two standard facts. (P1) If ζ is a simple loop, then $\mathrm{Push}(\zeta) = T_{\zeta_1} T_{\zeta_2}^{-1}$, where ζ_1, ζ_2 are the two boundary curves of a regular neighbourhood of ζ, the order being fixed by the orientation of ζ and the sign convention for twists [FM, Fact 4.7]. (P2) For every mapping class h and simple closed curve c, $hT_c h^{-1} = T_{h(c)}$ [FM, Fact 3.7].

Two consequences of (P1) are used. First, if $\Delta \subset \mathbb{P}^1$ is a closed disk with $\partial\Delta \cap B = \emptyset$ and $\Delta \cap B = \{v, w\}$, and ζ is a simple loop of v in $\Delta \setminus \{w\}$ going once around w, then one boundary curve of its regular neighbourhood bounds a disk containing only w (trivial twist) and the other is isotopic to $\partial\Delta$, so

$$\mathrm{Push}(\zeta) = T_{\partial\Delta}^{\pm 1}. \tag{2.3}$$

Second, if κ, η are disjoint simple closed curves bounding an annulus meeting B only in v, with κ inside η in a given disk picture, and ζ is a simple loop of v once counterclockwise around the core of the annulus, then $\mathrm{Push}(\zeta) = T_\eta^{\epsilon} T_\kappa^{-\epsilon}$ with $\epsilon = \pm 1$ depending only on the sign convention for Dehn twists.

Convention 2.1. We fix the sign convention for Dehn twists so that $\epsilon = +1$, that is, $\mathrm{Push}(\zeta) = T_\eta T_\kappa^{-1}$ for such a counterclockwise circuit. With the opposite convention the boxed formula (3.5) of Section 3 reads $T_\eta^{-2}(C)$; both curves have the stated property (Remark 7.4).

3 The model map and the curve

Let f be as in Section 1.2 and conjugate it so that its critical cycle is $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$ with 0 the critical point. A quadratic rational map with a critical point at 0 mapping to ∞ has a double pole at 0 and no other pole, so it is $1 + b_s/z + c_s/z^2$ after using $F(\infty) = 1$. Solving the linear equations $F(1) = s$ and $F(s) = 0$ for b_s, c_s gives $b_s = (-s^2 - s + 1)/(s - 1)$ and $c_s = s(2s - 1)/(s - 1)$, which is (1.3). The other critical point of F_s is $-2c_s/b_s$, and its critical value is $1 - b_s^2/(4c_s)$, which simplifies to

$$v(s) = -\frac{(s^2 - 3s + 1)^2}{4s(s - 1)(2s - 1)}. \tag{3.1}$$

The map has degree two and the four cycle points are distinct exactly when $s \neq 0, 1, 1/2$ (for $s = 1/2$ one has $c_s = 0$). The identities

$$v(s) - 1 = -\frac{(s^2 + s - 1)^2}{4s(s - 1)(2s - 1)}, \qquad v(s) - s = -\frac{(3s^2 - 3s + 1)^2}{4s(s - 1)(2s - 1)}$$

show that $v(s) \notin \{0, \infty, 1, s\}$, that is, $v(s)$ lies outside the cycle, exactly when the three quadratic factors below are nonzero. So the maps in the problem are, up to Möbius conjugacy, precisely the F_s with

$$s \in U = \left\{ s \in \mathbb{C} : s(s - 1)(2s - 1)(s^2 - 3s + 1)(s^2 + s - 1)(3s^2 - 3s + 1) \neq 0 \right\}. \tag{3.2}$$

The set U, a cofinite subset of $\mathbb{C}$, is connected. We write $A_s = \{0, \infty, 1, s\}$ and $B_s = A_s \cup \{v(s)\}$ for the marked sets of F_s. Our model is the parameter $a = 3/5$:

$$F_a(z) = 1 - \frac{1}{10z} - \frac{3}{10z^2}, \qquad 0 \longmapsto \infty \longmapsto 1 \longmapsto a \longmapsto 0, \tag{3.3}$$

with other critical point -6 and other critical value $v_a = v(3/5) = 121/120$. On the sphere marked by $B_a = \{0, \infty, 1, a, v_a\}$ draw the round circles

$$C = \left\{ |z| = \tfrac{241}{240} \right\}, \qquad \eta = \left\{ \left| z - \tfrac{4}{5} \right| = \tfrac{1}{4} \right\}. \tag{3.4}$$

The exterior of C contains exactly the marked points ∞, v_a (note $v_a = 242/240$), and the interior of η contains exactly $a, 1, v_a$ (the distances from $4/5$ to a, 1, v_a are $1/5$, $1/5$, $25/120$, all less than $1/4$, while $|0 - 4/5| > 1/4$). The two circles intersect. Our model curve is

$$\boxed{\gamma_a = T_\eta^2(C),} \tag{3.5}$$

with the sign of Convention 2.1.

Construction 3.1 (Transport to a general map)**.** *Let f be a map of the family with normalized form F_s, $s \in U$, the normalizing Möbius transformation being μ (so $\mu f \mu^{-1} = F_s$). Choose a path s_t, $0 \le t \le 1$, in U from a to s. The five points $0, \infty, 1, s_t, v(s_t)$ stay distinct, so by isotopy extension there is an ambient isotopy H_t of $\mathbb{P}^1$ with $H_0 = \mathrm{id}$ fixing $0, \infty, 1$ and with $H_t(a) = s_t$, $H_t(v_a) = v(s_t)$. Put $\gamma = \mu^{-1}(H_1(\gamma_a)) \subset \mathbb{P}^1 \setminus B$; equivalently, $H_1(\gamma_a)$ is the image of $H_1(C)$ under two twists about $H_1(\eta)$.*

The isotopy class of γ relative to B depends on the path s_t only through its homotopy class in U; Theorem 3.2 holds for every choice.

Theorem 3.2. *Let f, A, B be as in Section 1.2, and let γ be the curve of Construction 3.1. Then the Dehn twist T_γ preserves $\mathcal{D}_f$. If D_0 denotes the connected component of $\mathcal{D}_f$ containing the base point τ_0, then the components $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.*

Sections 4–7 prove the theorem for the model $f = F_a$ and $\gamma = \gamma_a$; Section 8 deduces the general case.

4 A finite cover of the moduli space

Recall Y and M from (2.1); M is the moduli space of B_a, with $0, \infty, 1$ normalized and (y, z) the positions of the points corresponding to a and v_a. A point of $\mathcal{T}(B_a)$ determines, through (1.1), a quadratic rational map with marked source and target points; the normalized positions of the fourth source and target points give a map from $\mathcal{T}(B_a)$ to a surface $\mathcal{W}$ of marked maps, a finite cover of M. This is the parameter surface of Hironaka and Koch [HKa, §§4.1–4.2]; in general such spaces are Hurwitz spaces, finite coverings of the moduli space of the target [Ko, Thm. 2.6], [KPS, §3.1].

Consider a quadratic rational map with marked source points $0, \infty, 1, x$ and marked target points $0, \infty, 1, y, z$ realizing the portrait

$$0 \overset{2}{\longmapsto} \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0, \tag{4.1}$$

where the 2 indicates a critical point, and having z as its other critical value. A degree-two map with a double pole at 0 and value 1 at ∞ is $(t-x)(t-r)/t^2$ with zeros x and r, and the condition $1 \mapsto y$ determines r. Hence the map is unique:

$$F_{x,y}(t) = \frac{(t-x)(t-r)}{t^2}, \qquad r = \frac{x+y-1}{x-1}. \tag{4.2}$$

Its critical points are 0 and $2xr/(x+r)$, and the critical value at the latter is $-(x-r)^2/(4xr)$, which in terms of x, y reads

$$Z(x,y) = \frac{((x-1)^2 - y)^2}{4x(x+y-1)(1-x)}. \tag{4.3}$$

The marked points must be distinct and the map must have degree two ($r \neq 0$), so the parameter surface and its projection to M are

$$\mathcal{W} = \{(x,y) \in Y^2 : x + y \neq 1,\ Z(x,y) \notin \{0, 1, y\}\}, \tag{4.4}$$

$$p\colon \mathcal{W} \longrightarrow M, \qquad p(x,y) = (y, Z(x,y)). \tag{4.5}$$

The maps F_s of the family (1.3) are the marked maps whose fourth source and target points agree: indeed $F_{s,s} = F_s$, the two expressions being quadratic maps with the same poles, zeros

and value at ∞, and $Z(s,s) = v(s)$. The conditions $Z(s,s) \neq 0, 1, s$ are precisely the three quadratic factors in (3.2), so the *diagonal* of $\mathcal{W}$ is

$$\mathcal{V} = \mathcal{W} \cap \{x = y\} = \{(s,s) : s \in U\}, \tag{4.6}$$

a connected Riemann surface, since U is connected.

Lemma 4.1. *1. The map p is an unramified covering of degree four, and $\mathcal{W}$ is connected.*

2. For the model (3.3), with $\pi_A \colon \mathcal{T}(A_a) \to Y$ and $\pi_B \colon \mathcal{T}(B_a) \to M$ the projections of Section 2, the map

$$q = (\pi_A \circ F_a^*,\ \pi_A \circ i_*) \colon \mathcal{T}(B_a) \longrightarrow \mathcal{W}$$

is well defined and satisfies $p \circ q = \pi_B$ and $q(\tau_0) = (a,a)$, and it is the universal covering of $\mathcal{W}$. Thus π_B factors through $\mathcal{W}$, and, with $\mathrm{pr}_x, \mathrm{pr}_y$ the two coordinate projections of $\mathcal{W}$, by construction

$$\pi_A \circ F_a^* = \mathrm{pr}_x \circ q, \qquad \pi_A \circ i_* = \mathrm{pr}_y \circ q. \tag{4.7}$$

Proof. (1) Fix $(y,z) \in M$. The equation $Z(x,y) = z$ is

$$((x-1)^2 - y)^2 + 4zx(x-1)(x+y-1) = 0,$$

a monic polynomial equation of degree four in x. Its values at $x = 0, 1, 1-y$ are $(1-y)^2$, y^2 and $y^2(y-1)^2$, all nonzero, so every root satisfies $x \in Y$ and $x + y \neq 1$; since $z \notin \{0, 1, y\}$, every root gives a point of $\mathcal{W}$. Thus $p^{-1}(y,z)$ consists of the roots of this equation. The roots are distinct because of the factorizations

$$\begin{aligned} Z - 1 &= -\frac{(x^2+y-1)^2}{4x(x-1)(x+y-1)}, \\ Z - y &= -\frac{(x^2+2xy-2x-y+1)^2}{4x(x-1)(x+y-1)}, \\ \frac{\partial Z}{\partial x} &= -\frac{(x^2+y-1)\,((x-1)^2-y)\,(x^2+2xy-2x-y+1)}{4x^2(x-1)^2(x+y-1)^2}, \end{aligned}$$

which show that $\partial Z/\partial x$ vanishes only where $Z = 1$, $Z = 0$ or $Z = y$, and these values of z are excluded in M. So p has exactly four preimages over each point, $\partial Z/\partial x \neq 0$ on $\mathcal{W}$, and the implicit function theorem shows that p is a local homeomorphism. A local homeomorphism between Hausdorff spaces all of whose fibres have the same finite cardinality is a covering (intersect the images of disjoint neighbourhoods of the four preimages of a point, and count), so p is a covering of degree four. Finally $\mathcal{W}$ is the complement in $\mathbb{C}^2$ of the finitely many algebraic curves listed in (4.4), and such a complement is path connected: join two points by two complex lines through a generic auxiliary point, each meeting the curves in finitely many points, which a path in the line can avoid.

(2) Let $\tau = [\phi] \in \mathcal{T}(B_a)$, with ϕ normalized to fix $0, \infty, 1$, and let $F \circ \psi = \phi \circ F_a$ be a pullback diagram (1.1) with ψ also normalized to fix $0, \infty, 1$ (possible since ψ is unique up

to a Möbius transformation). Put $x = \psi(a)$, $y = \phi(a)$, $z = \phi(v_a)$. Applying the diagram to the points of A_a and to the critical points $0, -6$ of F_a shows that F has the portrait (4.1) with marked source points $\{0, \infty, 1, x\}$, marked target points $\{0, \infty, 1, y, z\}$, and other critical value $F(\psi(-6)) = z$. Hence $F = F_{x,y}$ and $Z(x, y) = z$ by the uniqueness statement above; $x, y \in Y$ and $z \notin \{0, 1, y\}$ because ψ, ϕ are injective, and $x + y \neq 1$ because F has degree two. Therefore $(x, y) = (\pi_A F_a^* \tau, \pi_A i_* \tau) \in \mathcal{W}$ and $p(x, y) = (y, z) = \pi_B(\tau)$. For τ_0 we may take $\psi = \phi = \mathrm{id}$, so $q(\tau_0) = (a, a)$; and q is continuous because F_a^*, i_*, π_A are.

It remains to see that q is a covering. Let $N \subset M$ be an open ball evenly covered by both p and π_B, with $p^{-1}(N) = \bigsqcup_j N_j$ and $\pi_B^{-1}(N) = \bigsqcup_k \widetilde{N}_k$. The connected set $q(\widetilde{N}_k)$ lies in a single N_j, where $q|_{\widetilde{N}_k} = (p|_{N_j})^{-1} \circ \pi_B|_{\widetilde{N}_k}$ is a homeomorphism onto N_j. Hence each N_j is evenly covered by q (possibly with empty preimage), so $q(\mathcal{T}(B_a))$ is open and closed in the connected space $\mathcal{W}$, and q is a surjective covering; as $\mathcal{T}(B_a)$ is simply connected, it is the universal covering. □

5 When a loop preserves a component

We now set up the covering-space criterion of Hironaka and Koch [HK, §§2.8–2.11] in the coordinates of Lemma 4.1. Let $b = (a, a) = q(\tau_0)$. A loop g in $\mathcal{W}$ based at b determines, via the loop $p \circ g$ in M and (2.2), a deck transformation δ of π_B, which sends τ_0 to the endpoint τ_1 of the lift of g to $\mathcal{T}(B_a)$ starting at τ_0. It is also a deck transformation of q: $q \circ \delta$ and q are both lifts of π_B through p, and they agree at τ_0 because $q(\tau_1) = b$. We denote δ by the same letter g and write $g(\tau)$, as in Section 2. The two *coordinate loops* of g are the loops $\mathrm{pr}_x \circ g$ and $\mathrm{pr}_y \circ g$ in Y based at a.

Lemma 5.1. *Let $E = \mathrm{im}\big(\pi_1(\mathcal{V}, b) \to \pi_1(\mathcal{W}, b)\big)$ be the set of classes of loops in $\mathcal{W}$ that are homotopic in $\mathcal{W}$ to a loop in the diagonal $\mathcal{V}$. Let $g \in \pi_1(\mathcal{W}, b)$, acting on $\mathcal{T}(B_a)$ as above, and let D_0 be the connected component of $\mathcal{D}_{F_a}$ containing τ_0.*

1. *If the two coordinate loops of g have equal classes in $\pi_1(Y, a)$, then $g(\mathcal{D}_{F_a}) = \mathcal{D}_{F_a}$.*
2. *$g(D_0) = D_0$ if and only if $g \in E$.*

Consequently, if the coordinate loops of g are homotopic and $g^n \notin E$ for every $n \neq 0$, then g preserves $\mathcal{D}_{F_a}$ and the components $g^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.

Proof. (1) By (4.7), F_a^* and i_* are lifts through the universal covering $\pi_A \colon \mathcal{T}(A_a) \to Y$ of the maps $\mathrm{pr}_x \circ q$ and $\mathrm{pr}_y \circ q$, and they take the same value at τ_0 because $\tau_0 \in \mathcal{D}_{F_a}$. Since g is a deck transformation of q, the map $F_a^* \circ g$ is another lift of $\mathrm{pr}_x \circ q$, so $F_a^* \circ g = \delta_x \circ F_a^*$ for a deck transformation δ_x of π_A. To identify δ_x, lift g to a path $\tilde{g}$ from τ_0 to $g(\tau_0)$; then $F_a^* \circ \tilde{g}$ is the lift, starting at $F_a^*(\tau_0)$, of the loop $\mathrm{pr}_x \circ g$, so δ_x is the deck transformation of π_A determined by the class of $\mathrm{pr}_x \circ g$ relative to the base point $F_a^*(\tau_0)$. Likewise $i_* \circ g = \delta_y \circ i_*$ with δ_y determined by the class of $\mathrm{pr}_y \circ g$ relative to the same base point $i_*(\tau_0) = F_a^*(\tau_0)$. Equal classes give $\delta_x = \delta_y =: \delta$, and then for $\tau \in \mathcal{D}_{F_a}$,

$$F_a^*(g\tau) = \delta F_a^*(\tau) = \delta\, i_*(\tau) = i_*(g\tau),$$

so $g(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$. The coordinate loops of g^{-1} are the inverses of those of g, hence also homotopic, so $g^{-1}(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$ as well.

(2) First, $\mathcal{D}_{F_a} \subset q^{-1}(\mathcal{V})$: if $F_a^*\tau = i_*\tau$, the two coordinates of $q(\tau)$ agree by (4.7). Next, on a connected component N of $q^{-1}(\mathcal{V})$, the maps $F_a^*|_N$ and $i_*|_N$ are lifts through π_A of one and the same map $\tau \mapsto \mathrm{pr}_x\, q(\tau) = \mathrm{pr}_y\, q(\tau)$, so by uniqueness of lifts they agree on all of N if they agree at one point. Hence $\mathcal{D}_{F_a}$ is a union of connected components of $q^{-1}(\mathcal{V})$; these are path connected, $q\colon q^{-1}(\mathcal{V}) \to \mathcal{V}$ being a covering of the connected manifold $\mathcal{V}$, and D_0 is the component through τ_0.

The deck transformation g permutes the components of $q^{-1}(\mathcal{V})$, so $g(D_0) = D_0$ if and only if there is a path in D_0 from τ_0 to $g(\tau_0)$. Given such a path, its image under q is a loop in $\mathcal{V}$ based at b, homotopic in $\mathcal{W}$ to g because both lift to paths in the simply connected space $\mathcal{T}(B_a)$ with the same endpoints; so $g \in E$. Conversely, if g is homotopic in $\mathcal{W}$ to a loop g' in $\mathcal{V}$, the lift of g' starting at τ_0 stays in $q^{-1}(\mathcal{V})$, hence in D_0, and ends at $g(\tau_0)$ by homotopy lifting; so $g(\tau_0) \in D_0$.

Finally, if $g^j(D_0) = g^k(D_0)$ with $j \neq k$, then $g^{j-k}(D_0) = D_0$, so $g^{j-k} \in E$ by (2), contrary to hypothesis. □

6 Loops in $\mathcal{W}$ representing the twist about γ_a

The goal of this section is a loop g in $\mathcal{W}$ with $P(p \circ g) = T_{\gamma_a}^{\pm 1}$ and trivial coordinate loops (Proposition 6.3). We work in the fibre $\mathcal{W}_a = \{x \in \mathbb{C} : (x, a) \in \mathcal{W}\}$ over $y = a = 3/5$, along which p fixes the four points of A_a and moves only v_a, so that mapping classes are point pushes of v_a. By (4.4) and the factorizations in the proof of Lemma 4.1, the points of $\mathbb{C}$ excluded from $\mathcal{W}_a$ are

$$0,\ 1,\ \tfrac{2}{5}, \qquad 1 \pm \sqrt{\tfrac{3}{5}}\ \ (Z = 0), \qquad \pm\sqrt{\tfrac{2}{5}}\ \ (Z = 1), \qquad \tfrac{2}{5} \pm \tfrac{i\sqrt{6}}{5}\ \ (Z = a); \tag{6.1}$$

here $2/5 = 1 - a$ is the pole of $Z(\cdot, a)$. Their moduli are 0, 1, 0.4, 1.77…, 0.22…, and $\sqrt{2/5} = 0.632\ldots$ for the last four; none is $a = 0.6$, so $\ell(\theta) = (ae^{2\pi i\theta}, a)$, $0 \le \theta \le 1$, is a loop in $\mathcal{W}$ based at b. Its image $p \circ \ell$ in M moves v_a along the loop $\zeta(\theta) = Z(ae^{2\pi i\theta}, a)$ in $\mathbb{P}^1 \setminus A_a$ based at $v_a = Z(a, a)$.

Let m be a positive meridian about the pole $x = 2/5$ in $\mathcal{W}_a$: travel from a along the real interval toward $2/5$, go once counterclockwise around a small circle centred at $2/5$, and return along the interval.

Lemma 6.1. *The mapping class $P(p \circ m)$ is $T_C^{\pm 1}$, a single Dehn twist about the round circle C of* (3.4).

Proof. On $(2/5, 3/5]$ the factorizations give $\partial Z(x, a)/\partial x < 0$, because in the numerator of $\partial Z/\partial x$ the factors $x^2 - 2/5$ and $(x-1)^2 - 3/5$ are negative and $x^2 - \frac{4}{5}x + \frac{2}{5}$ is positive. Moreover $Z(a, a) = v_a$, so $Z(x, a) \ge v_a > 1$ on $(2/5, a]$ by monotonicity; the pole of $Z(\cdot, a)$ at $2/5$ is simple, and $Z(x, a) \to +\infty$ as $x \downarrow 2/5$. Thus $p \circ m$ moves v_a out along the positive real ray, once around ∞ (a small circle about a simple pole maps to a loop winding once

about ∞), and back. For a small enough circle this motion takes place in the closed disk $\Delta = \{|z| \geq 241/240\} \cup \{\infty\}$ bounded by C, whose only marked points are v_a and ∞; in $\Delta \setminus \{\infty\}$, whose fundamental group is $\mathbb{Z}$, the loop is homotopic to a simple loop once around ∞, so (2.3) gives $P(p \circ m) = T_C^{\pm 1}$. $\square$

The homotopy class of ζ, not merely its homology class, must be identified, since the isotopy class of γ_a depends on it.

Lemma 6.2. *The loop ζ is homotopic, as a loop based at v_a in $\mathbb{P}^1 \setminus A_a$, to two clockwise circuits around the segment $[a,1]$ inside the disk bounded by η. Consequently $h := P(\zeta^{-1})$ satisfies $h(C) = T_\eta^2(C) = \gamma_a$, with the sign of Convention 2.1.*

Proof. Step 1: enlarging the circle. Put $R = \sqrt{1-a} = \sqrt{2/5} > a$. No excluded point in (6.1) has modulus in $[a, R)$, so for $a \leq r < R$ the loops $\zeta_r(\theta) = Z(re^{2\pi i\theta}, a)$ avoid $0, a, 1, \infty$ and depend continuously on r. Their base points $Z(r,a)$ move along the real path $\sigma_r = Z([a,r],a) \subset (1, v_a]$, since $Z(\cdot, a)$ is decreasing on $(2/5, R)$ (the sign analysis in the proof of Lemma 6.1 is valid there) with $Z(R,a) = 1$. Hence ζ is based-homotopic in $\mathbb{P}^1 \setminus A_a$ to $\sigma_r^{-1}\zeta_r\sigma_r$ for every $r \in [a, R)$.

Step 2: the circle $|x| = R$ maps onto $[a,1]$, and nothing inside it does. The substitution $u = x + (1-a)/x$ (a Joukowski map) simplifies Z: writing $(x-1)^2 - a = x(u-2)$ and $(x+a-1)(1-x) = x(2-a-u)$ gives

$$Z(x,a) = G(u), \qquad G(u) = \frac{(u-2)^2}{4(2-a-u)}. \tag{6.2}$$

For $x = Re^{i\vartheta}$ we have $u = 2R\cos\vartheta$, so on $|x| = R$ the variable u runs over $[-2R, 2R]$, where $2 - a - u \geq 2 - a - 2R > 0$ and

$$G(u) - 1 = \frac{u^2 - 4(1-a)}{4(2-a-u)}, \qquad G(u) - a = \frac{(u - 2(1-a))^2}{4(2-a-u)}.$$

Thus on $[-2R, 2R]$ the function G takes values in $[a,1]$, equals 1 exactly at the endpoints, and equals a exactly at $u = 2(1-a)$; since $G'(u) = (u-2)(2-2a-u)/(4(2-a-u)^2)$, G decreases from 1 to a and then increases back to 1. Hence for $z \in (a,1)$ the equation $G(u) = z$ has two distinct solutions $u \in (-2R, 2R)$, and for each of them the two roots of $x^2 - ux + (1-a) = 0$ are non-real conjugates with $|x|^2 = 1 - a$, hence lie on $|x| = R$. This produces four distinct points of $|x| = R$ mapping to z, which exhaust the four preimages of (a,z) under p. For the endpoints the factorizations of $Z - 1$ and $Z - a$ show that all preimages of 1 and of a (namely $\pm R$ and $2/5 \pm i\sqrt{6}/5$, with multiplicity) also lie on $|x| = R$. So $Z(\{|x| = R\}, a) = [a,1]$, and no x with $|x| < R$ maps into $[a,1]$.

Step 3: the homotopy class. The segment $[a,1]$ lies in the open disk bounded by η, so by Step 2 and continuity, for $r < R$ close to R the loop ζ_r lies in that disk and avoids $[a,1]$; the base point path $\sigma_r \subset (1, v_a]$ also lies in the disk and avoids $[a,1]$. So $\sigma_r^{-1}\zeta_r\sigma_r$ is a loop in the η-disk minus the segment $[a,1]$. This space is an annulus, with fundamental group $\mathbb{Z}$ generated by a counterclockwise circuit around $[a,1]$, and the class of a loop is its winding

number about any point of the segment, say about 1. Since the homotopy of Step 1 avoids 1, this winding number equals $\mathrm{wind}_1(\zeta)$, which we compute by the argument principle applied to

$$Z(x,a) - 1 = -\frac{(x^2 + a - 1)^2}{4x(x-1)(x+a-1)}$$

on the disk $|x| < a$: the numerator vanishes only at $\pm R$, outside the disk, while the denominator has the two simple zeros 0 and $1 - a = 2/5$ inside it. Hence $\mathrm{wind}_1(\zeta) = 0 - 2 = -2$, and ζ is based-homotopic to two clockwise circuits around $[a, 1]$ in the η-disk, as claimed.

Step 4: the point push. Choose a round circle κ enclosing $a, 1$ but not v_a and lying inside both C and η; for instance $\kappa = \{|z - \frac{4}{5}| = \frac{97}{480}\}$, for which $|a - \frac{4}{5}| = |1 - \frac{4}{5}| = \frac{96}{480} < \frac{97}{480} < \frac{100}{480} = |v_a - \frac{4}{5}|$, and the largest modulus on κ is $\frac{481}{480} < \frac{241}{240}$. The annulus between κ and η contains the single marked point v_a, and by Step 3, ζ^{-1} is homotopic to two counterclockwise circuits of v_a around this annulus. By Convention 2.1 one such circuit has point push $T_\eta T_\kappa^{-1}$, so, the twists about the disjoint curves κ and η commuting,

$$h = P(\zeta^{-1}) = T_\eta^2 T_\kappa^{-2}.$$

Since κ is disjoint from C, the twist T_κ fixes the isotopy class of C, and $h(C) = T_\eta^2(C) = \gamma_a$. □

Proposition 6.3. *Let*

$$g = \ell^{-1} m \ell \in \pi_1(\mathcal{W}, b), \tag{6.3}$$

the loop that traverses ℓ, then m, then ℓ backwards. Then $P(p \circ g) = T_{\gamma_a}^{\pm 1}$: the deck transformation g acts on $\mathcal{T}(B_a)$ as the Dehn twist about γ_a or its inverse. Both coordinate loops of g are null-homotopic in Y. In particular T_{γ_a} preserves $\mathcal{D}_{F_a}$.

Proof. Since $p \circ \ell$ moves only v_a, along ζ, we have $P(p \circ \ell) = P(\zeta) = h^{-1}$ with h as in Lemma 6.2. Multiplicativity of P (Section 2) and Lemma 6.1 give

$$P(p \circ g) = P(p \circ \ell^{-1})\, P(p \circ m)\, P(p \circ \ell) = h\, T_C^{\pm 1}\, h^{-1} = T_{h(C)}^{\pm 1} = T_{\gamma_a}^{\pm 1},$$

using (P2) and Lemma 6.2. The y-coordinate of g is constant. The x-coordinate of m is a small loop around $2/5$ together with a real approach path in $(2/5, 3/5]$; this loop lies in the disk $|x - \frac{1}{2}| < \frac{1}{2}$, which avoids 0 and 1, so it is null-homotopic in Y; hence the x-coordinate loop of g is homotopic to $(\mathrm{pr}_x \circ \ell)^{-1}(\mathrm{pr}_x \circ \ell)$, which is trivial. Lemma 5.1(1) now shows that g, hence T_{γ_a}, preserves $\mathcal{D}_{F_a}$. □

7 No nonzero power of the twist preserves the base component

By Lemma 5.1 and Proposition 6.3, it remains to show that $g^n \notin E$ for $n \neq 0$. Consider

$$\Phi \colon \mathcal{W} \longrightarrow Y \times \mathbb{C}^*, \qquad \Phi(x,y) = \big(t(x,y), L(x,y)\big) = \left(\frac{xy}{(1-x)(1-y)},\ (1-x)(1-y)\right). \tag{7.1}$$

It is well defined: $L \neq 0$ and $t \neq 0$ since $x, y \in Y$, and $t-1 = (x+y-1)/((1-x)(1-y)) \neq 0$. In terms of the symmetric functions $e_1 = x+y$, $e_2 = xy$ one has $L = 1-e_1+e_2$ and $t = e_2/L$, so Φ is the symmetric quotient $(x, y) \mapsto (x+y, xy)$ of Hironaka–Koch [HK, §2.12], restricted to $\mathcal{W}$, up to this change of coordinates.

The constraint on diagonal loops. Enlarge the diagonal by filling in its six quadratic punctures: $\widehat{\mathcal{V}} = \{(s, s) : s \in \mathbb{C} \setminus \{0, 1, 1/2\}\} \supset \mathcal{V}$, on which (7.1) still makes sense, so that the covering (7.3) below covers all of Y. In the coordinate $w = s/(1-s)$, which maps $\mathbb{C} \setminus \{0, 1, 1/2\}$ bijectively onto $\mathbb{C} \setminus \{0, 1, -1\}$, the restriction of Φ is

$$t = w^2, \qquad L = (1+w)^{-2}, \qquad w \in \mathbb{C} \setminus \{0, 1, -1\}, \tag{7.2}$$

because $1 + w = 1/(1-s)$. The squaring map

$$\mathbb{C} \setminus \{0, 1, -1\} \longrightarrow Y, \qquad w \longmapsto w^2, \tag{7.3}$$

is a two-sheeted covering; our base points are $w_0 = a/(1-a) = 3/2$ and $t_0 = w_0^2 = 9/4$.

Lemma 7.1. *Let λ be a loop in $\mathcal{V}$ based at b. Then $w \circ \lambda$ is the lift of the loop $t \circ \lambda$ through* (7.3) *starting at w_0; it is closed, and*

$$\operatorname{wind}_0(L \circ \lambda) = -2 \operatorname{wind}_{-1}(w \circ \lambda). \tag{7.4}$$

The based homotopy class of $w \circ \lambda$ in $\mathbb{C} \setminus \{0, 1, -1\}$, hence the right-hand side of (7.4)*, depends only on the based homotopy class of $t \circ \lambda$ in Y.*

Proof. On $\mathcal{V} \subset \widehat{\mathcal{V}}$ formula (7.2) applies: $w \circ \lambda$ is a closed path in $\mathbb{C} \setminus \{0, 1, -1\}$ from w_0 with $(w \circ \lambda)^2 = t \circ \lambda$, hence the lift of $t \circ \lambda$ from w_0, and $L \circ \lambda = (1 + w \circ \lambda)^{-2}$ gives (7.4). The last assertion is homotopy lifting for the covering (7.3). □

The loops g^n violate the constraint. On the fibre $y = a$ the map (7.1) becomes

$$t = \frac{3x}{2(1-x)}, \qquad L = \frac{2}{5}(1-x). \tag{7.5}$$

Let $\alpha = t \circ \ell$ and $\beta = t \circ m$, loops in Y based at t_0.

Lemma 7.2. *For every integer n, the loop g^n satisfies*

$$\operatorname{wind}_0(L \circ g^n) = 0, \qquad \operatorname{wind}_{-1}\big(\widetilde{t \circ g^n}\big) = n, \tag{7.6}$$

where $\widetilde{t \circ g^n}$ is the lift of $t \circ g^n$ through (7.3) *starting at w_0, which is a closed loop.*

Proof. The loop ℓ. Along ℓ we have $x = ae^{2\pi i\theta}$ with $|x| = a < 1$, so $1-x$ does not wind about 0 and (7.5) gives $\operatorname{wind}_0(\alpha) = 1$ and $\operatorname{wind}_0(L \circ \ell) = 0$. Since $\operatorname{wind}_0(\alpha)$ is odd, continuation of the square root along α, or along α^{-1}, interchanges the two sheets of (7.3): the lift $\tilde{\alpha}$ of α starting at w_0 ends at $-w_0$.

The loop m. The Möbius map $x \mapsto t$ in (7.5) sends $2/5$ to 1 and a to $9/4$, is increasing on the real interval $(2/5, a]$, which it maps onto $(1, 9/4]$, and, being conformal, sends a small counterclockwise circle about $2/5$ to a small counterclockwise loop about 1. Hence β is a positive meridian about 1 in Y, approached from t_0 along the real axis to the right of 1. Its lift starting at $w_0 = 3/2$ runs along $(1, 3/2]$ and around a small loop about $+1$, so it is a positive meridian about $+1$; the lift starting at $-w_0$ is its negative, a positive meridian about -1 (the map $w \mapsto -w$ preserves orientation). Also $\mathrm{wind}_0(L \circ m) = 0$, since $1 - x$ stays near the positive interval $[2/5, 3/5]$ along m.

The loop g. By additivity, $\mathrm{wind}_0(L \circ g) = -\mathrm{wind}_0(L \circ \ell) + \mathrm{wind}_0(L \circ m) + \mathrm{wind}_0(L \circ \ell) = 0$, and so $\mathrm{wind}_0(L \circ g^n) = 0$. The t-projection of g is $\alpha^{-1}\beta\alpha$: first α, then β, then α backwards. Its lift from w_0 consists of $\tilde{\alpha}$, ending on the other sheet at $-w_0$, then the lift of β from $-w_0$, a positive meridian about -1, then $\tilde{\alpha}$ reversed, returning to w_0. The contributions of $\tilde{\alpha}$ and its reverse to the change of argument of $w + 1$ cancel, so this closed lift has winding number 1 about -1, and the lift of $t \circ g^n = (\alpha^{-1}\beta\alpha)^n$ from w_0, the n-fold concatenation, has winding number n. □

Proposition 7.3. *For every $n \neq 0$, $g^n \notin E$. Consequently Theorem 3.2 holds for the model map F_a and the curve $\gamma_a = T_\eta^2(C)$.*

Proof. Suppose $g^n \in E$, so that g^n is homotopic in $\mathcal{W}$ to a loop λ in $\mathcal{V}$ based at b. Applying Φ, the loops $t \circ \lambda$ and $t \circ g^n$ are based-homotopic in Y, and $L \circ \lambda$ and $L \circ g^n$ are homotopic in $\mathbb{C}^*$. By Lemma 7.1, the lift $w \circ \lambda$ is based-homotopic to the lift of $t \circ g^n$ used in Lemma 7.2, so $\mathrm{wind}_{-1}(w \circ \lambda) = n$, while $\mathrm{wind}_0(L \circ \lambda) = \mathrm{wind}_0(L \circ g^n) = 0$. The constraint (7.4) then reads $0 = -2n$, so $n = 0$.

For the second assertion, g acts as $T_{\gamma_a}^{\pm 1}$ and has homotopic coordinate loops (Proposition 6.3), so by Lemma 5.1 the twist T_{γ_a} preserves $\mathcal{D}_{F_a}$ and the components $T_{\gamma_a}^n(D_0) = g^{\pm n}(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct. □

Remark 7.4. The same computation applies to the other conjugate $g' = \ell m \ell^{-1}$ (its t-projection $\alpha\beta\alpha^{-1}$ also changes sheet before the meridian, as α^{-1} has odd winding number about 0, and again $\mathrm{wind}_0(L \circ g') = 0$), whose mapping class is $h^{-1}T_C^{\pm 1}h = T_{h^{-1}(C)}^{\pm 1}$ with $h^{-1}(C) = T_\eta^{-2}T_\kappa^2(C) = T_\eta^{-2}(C)$. Hence Theorem 3.2 holds for both curves $T_\eta^{\pm 2}(C)$.

8 Transport to an arbitrary map of the family

Let f, μ, F_s, s_t, H_t be as in Construction 3.1. We show that changing markings by $H = H_1$ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$, D_0 to the base component of $\mathcal{D}_{F_s}$, and conjugates T_{γ_a} to $T_{H(\gamma_a)}$.

Lemma 8.1. *There is a continuous family of homeomorphisms K_t of $\mathbb{P}^1$, $0 \le t \le 1$, with $K_0 = \mathrm{id}$ and*

$$F_{s_t} \circ K_t = H_t \circ F_a \qquad (0 \le t \le 1). \tag{8.1}$$

It satisfies $K_t|_{A_a} = H_t|_{A_a}$ for all t; consequently $H_t^{-1}K_t$ is an isotopy from the identity through homeomorphisms fixing A_a pointwise, and K_1 is isotopic to H_1 relative to A_a.

Proof. Every F_s is Möbius-equivalent to $Q(z) = z^2$, with Möbius maps depending continuously on s: completing the square in $F_s(z) = 1 + b_s/z + c_s/z^2$ and recalling $v(s) = 1 - b_s^2/(4c_s)$ gives $F_s = v(s) + c_s\, J_s^2$, that is,

$$\nu_s \circ F_s = Q \circ J_s, \qquad J_s(z) = \frac{1}{z} + \frac{b_s}{2c_s}, \qquad \nu_s(w) = \frac{w - v(s)}{c_s},$$

with $c_s \neq 0$ for $s \in U$. The isotopy $\widehat{H}_t = \nu_{s_t} \circ H_t \circ \nu_a^{-1}$ starts at the identity and fixes 0 and ∞, because $H_t(v_a) = v(s_t)$ and $H_t(\infty) = \infty$. By the homotopy lifting property of the covering $Q\colon \mathbb{C}^* \to \mathbb{C}^*$, the homotopy $(z,t) \mapsto \widehat{H}_t(Q(z))$ lifts uniquely to a continuous $\rho\colon \mathbb{C}^* \times [0,1] \to \mathbb{C}^*$ with $\rho_0 = \mathrm{id}$ and $Q \circ \rho_t = \widehat{H}_t \circ Q$; lifting $\widehat{H}_t^{-1} \circ Q$ in the same way gives ρ_t' with $\rho_t'\rho_t = \mathrm{id} = \rho_t\rho_t'$ (both products lift the constant homotopy Q and start at id), so each ρ_t is a homeomorphism of $\mathbb{C}^*$, and ρ extends continuously to $\mathbb{P}^1 \times [0,1]$ by $\rho_t(0) = 0$, $\rho_t(\infty) = \infty$, since $|\rho_t(z)|^2 = |\widehat{H}_t(z^2)|$ and $\widehat{H}$ is continuous on the compact space $\mathbb{P}^1 \times [0,1]$. Then $K_t = J_{s_t}^{-1} \circ \rho_t \circ J_a$ is a continuous family of homeomorphisms with $K_0 = \mathrm{id}$ and $F_{s_t}K_t = \nu_{s_t}^{-1} Q \rho_t J_a = \nu_{s_t}^{-1} \widehat{H}_t Q J_a = H_t \nu_a^{-1} Q J_a = H_t F_a$, which is (8.1).

For the marked points, (8.1) gives $F_{s_t}(K_t(w)) = H_t(F_a(w))$ for $w \in A_a$. Thus $K_t(0)$ is the critical point of F_{s_t} over ∞, namely 0; $K_t(\infty)$ lies in $F_{s_t}^{-1}(1) = \{\infty, -c_{s_t}/b_{s_t}\}$, $K_t(1)$ in $F_{s_t}^{-1}(s_t)$, and $K_t(a)$ in $F_{s_t}^{-1}(0) \ni s_t$. For $s_t \in U$ each of the latter three fibres (over the non-critical values 1, s_t, 0) consists of two distinct points depending continuously on t: $b_{s_t} \neq 0$ since $s_t^2 + s_t - 1 \neq 0$, and s_t, 0 are not critical values of F_{s_t} since $v(s_t) \neq s_t, 0$. By continuity from $t = 0$, $K_t(0) = 0$, $K_t(\infty) = \infty$, $K_t(1) = 1$ and $K_t(a) = s_t$, which are the values of H_t on A_a. The remaining assertions follow. □

Proof of Theorem 3.2. Write $H = H_1$, $K = K_1$. Since H maps B_a onto B_s (notation of Section 3), changing markings by H is a biholomorphic isomorphism

$$\Psi\colon \mathcal{T}(B_a) \longrightarrow \mathcal{T}(B_s), \qquad [\phi] \longmapsto [\phi \circ H^{-1}],$$

and likewise $[\psi] \mapsto [\psi \circ H^{-1}]$ identifies $\mathcal{T}(A_a)$ with $\mathcal{T}(A_s)$, $A_s = \{0, \infty, 1, s\} = H(A_a) = K(A_a)$.

Ψ *carries* $\mathcal{D}_{F_a}$ *onto* $\mathcal{D}_{F_s}$. Let ϕ be a marking with pullback diagram $\widetilde{F} \circ \psi = \phi \circ F_a$ for F_a. Using $F_a \circ K^{-1} = H^{-1} \circ F_s$ from (8.1),

$$\widetilde{F} \circ (\psi K^{-1}) = \phi \circ F_a \circ K^{-1} = (\phi H^{-1}) \circ F_s,$$

so ψK^{-1} is a source marking for the target marking ϕH^{-1} in a pullback diagram for F_s: $F_s^*\Psi[\phi] = [\psi K^{-1}]$, while $i_*\Psi[\phi] = [\phi H^{-1}]$. Now $[\psi K^{-1}] = [\phi H^{-1}]$ in $\mathcal{T}(A_s)$ if and only if $[\psi K^{-1} H] = [\phi]$ in $\mathcal{T}(A_a)$, and $K^{-1}H$ is isotopic to the identity relative to A_a by Lemma 8.1, so this holds if and only if $[\psi] = [\phi]$ in $\mathcal{T}(A_a)$, that is, $[\phi] \in \mathcal{D}_{F_a}$. Hence $\Psi(\mathcal{D}_{F_a}) = \mathcal{D}_{F_s}$.

Ψ *conjugates twists.* For a mapping class $k \in \mathrm{PMod}(B_a)$, $\Psi([\phi] \cdot k) = [\phi k H^{-1}] = [\phi H^{-1}] \cdot (HkH^{-1})$, so Ψ intertwines the action of T_{γ_a} with that of $HT_{\gamma_a}H^{-1} = T_{H(\gamma_a)}$, by the naturality of twists (P2), which holds verbatim for the homeomorphism $H\colon (\mathbb{P}^1, B_a) \to (\mathbb{P}^1, B_s)$.

Ψ *carries* D_0 *to the base component.* The path of markings $[H_t]$ in $\mathcal{T}(B_a)$ lies in $\mathcal{D}_{F_a}$: by (8.1), K_t is a source marking for the target marking H_t, and $[K_t] = [H_t]$ in $\mathcal{T}(A_a)$ because $H_t^{-1}K_t$ is isotopic to the identity relative to A_a (Lemma 8.1). The path starts at τ_0, so its endpoint $[H]$ lies in D_0, and $\Psi[H] = [\mathrm{id}]$ is the base point of $\mathcal{T}(B_s)$. Hence $\Psi(D_0)$ is the base component of $\mathcal{D}_{F_s}$, and the components $T^n_{H(\gamma_a)}(\Psi(D_0)) = \Psi(T^n_{\gamma_a}(D_0))$, $n \in \mathbb{Z}$, are pairwise distinct by Proposition 7.3. Finally, conjugating by μ^{-1} identifies $(\mathbb{P}^1, B_s, F_s)$ with $(\mathbb{P}^1, B, f)$ and $H(\gamma_a)$ with γ, which proves the theorem for f. □

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